

Multi Agent Class

Quick Reference Card — Links, Commands & Submission Guide

Feel free to mail me at s25089@students.iitmandi.ac.in (YASHS TIWARI)

Credentials (all machines)

SSH username: teaching | Password: ds123 | Conda env: agentic | Project folder: ~/nexaai/project/

First Thing Every Session — Activate Environment

```
# Open a terminal in code-server (Ctrl+`) then run:
conda activate agentic
cd ~/nexaai/project

# Verify RAG is reachable:
python3 -c "from global_rag_client import verify_rag_connection;
verify_rag_connection()"
```

How Each Department Tests (No Penalty — Unlimited)

Department team members NEVER run main.py. They test their own agent file directly. No submission counter, no penalty — run as many times as needed.

STEP 1 — Add this block at the bottom of your agent file:

```
# Add at the bottom of agents/finance_agent.py (or your dept file)
if __name__ == '__main__':
    import json
    agent = FinanceAgent()          # use your agent class name
    agent.analyze('Test scenario')
    output = agent.get_output()
    print(json.dumps(output, indent=2, default=str))
```

STEP 2 — Run your agent directly:

```
python3 agents/finance_agent.py      # Finance
python3 agents/engineering_agent.py  # Engineering
python3 agents/marketing_agent.py    # Marketing
python3 agents/hr_agent.py           # HR
python3 agents/legal_agent.py        # Legal
```

You will see your full JSON output in the terminal. Fix errors and run again. No limit.

How CEO Team Tests and Submits

CEO team runs main.py. Two modes:

Command	What It Does	Penalty ?	When to Use
python3 main.py --validate-only	Checks schema, shows errors grouped by category. Does NOT submit. Does NOT increment counter.	NONE — unlimited	Use this constantly. Fix all errors here first.
python3 main.py	Full run: reads RAG, runs all agents, validates schema, writes submission JSON to submissions/.	YES after 1st submission	Only when --validate-only passes cleanly.

Deployment Penalty — Read Before Submitting

1st submission: 0% penalty (free)

2nd submission: -5% on final score

3rd submission: -12% | 4th: -22% | 5th+: -35%

Run --validate-only until it passes. Then submit once with confidence.

What Happens Automatically When You Run main.py

Step	What Happens	If It Fails
1	Connects to Global RAG at 172.18.40.114:3000	Stops immediately — check WiFi / RAG server
2	Reads all RAG files and computes SHA256 hash	Cached version used if temporary failure
3	Reads active scenario (H0/H1/H2/H3) and market parameters	Uses last known values
4	Runs all 5 dept agents in order (HR→Eng→Finance→Legal→Mkt)	Error shown, fix the agent
5	Validates full schema — 8 categories of checks	Shows grouped errors with hints. Fix and re-run.
6	Writes submission_team_X_NN_HHMMSS.json to submissions/	File system error — check disk space

The RAG hash check is automatic: if your agent hardcoded any market parameter instead of reading from RAG, the hash will not match and you receive a staleness penalty. Always use `get_market_param()` from `global_rag_client`.

Scoring Process — What Happens After You Submit

Step	Who	What
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1	CEO team	Run python3 main.py — submission JSON written to submissions/
2	CEO team	Tell the instructor: 'team_X submitted'
3	Instructor	SSH into your machine, copy the JSON file
4	Instructor	Shares JSON with evaluator — scores calculated manually
5	Instructor	Enters scores in scoreboard control panel → Push to RAG
6	All teams	Scores appear on the live scoreboard automatically

You will NOT see your scores immediately after running main.py. Scores appear on the scoreboard ONLY after the instructor pushes them. Wait for the announcement.

Quick Troubleshooting

Error	Fix
Cannot connect to RAG server	conda activate agentic first. Check config.py: RAG_SERVER_URL = "http://172.18.40.114:3000"
ModuleNotFoundError	conda activate agentic — you are in wrong Python environment
Schema validation failed	Read the error groups carefully. Each group has a hint. Fix one group at a time.
tools_called is empty	Use self.call_tool(self.my_function, ...) — not self.my_function(...) directly
confidence_method rejected	Must describe HOW you calculated confidence — not 'LLM estimated'
stated_reasoning too short	Must be ≥ 100 chars AND cite tool names and their output values
Ollama not responding	The agentic conda env activates Ollama automatically. Re-run: conda activate agentic

```
# If anything breaks, restart clean:
conda activate agentic
cd ~/nexaai/project
python3 main.py --validate-only
```